ADITYA BHOITE

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Education

Vidyalankar Institute of Technology, Mumbai 2022 – 2026
- B.Tech in Electronics & Computer Science | *CGPA*: **8.87**

PACE Junior Science College, Kalyan 2020 – 2022
- *HSC (Class XII)*

Skills

Languages: Python, C++, SQL, JavaScript
Libraries/Frameworks: Scikit-learn, TensorFlow/Keras, XGBoost, NumPy, Pandas, Sentence Transformers, Matplotlib, Seaborn, Streamlit
AI/ML Concepts: Machine Learning, Deep Learning (CNN), Supervised/Unsupervised Learning, Model Evaluation & Benchmarking, LLMs, Prompt Engineering, RAG, Text Embeddings, Semantic Search
Data: Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Data Preprocessing, Statistical Analysis
Tools: Git/GitHub, Docker, AWS, PostgreSQL, Google Colab, Jupyter Notebook, Power BI

Projects

DevResolve — RAG-Based Developer Troubleshooting System (LLM) GitHub

A RAG-based LLM system that generates evidence-grounded answers from technical documentation.
- Built a **Retrieval-Augmented Generation (RAG)** pipeline in **Python** that uses a curated technical documentation corpus to generate **source-cited troubleshooting explanations**.
- Implemented document retrieval using **Sentence Transformers embeddings** for semantic search and **BM25 keyword search**, and combined them into a **hybrid retrieval** approach.
- Applied **prompt engineering** to ground LLM responses in retrieved documentation and to handle questions where sufficient evidence was unavailable.
- Created a **51-question evaluation set** covering answerable and unsupported queries; ran experiments to benchmark chunk sizes and **vector versus hybrid retrieval**.

Financial Resilience Intelligence GitHub

Financial resilience analysis of 25,000+ US households using Federal Reserve survey data (2024–2025).
- Cleaned survey microdata and built a transparent financial resilience score, validated against an independent Fed measure ($p < 0.001$, **Cohen's d = 1.48**).
- Engineered financial-behavior features, improving **Logistic Regression ROC-AUC from 0.799 to 0.867**; benchmarked against **Random Forest** (0.891) using **scikit-learn**.
- Investigated a year-over-year resilience decline in low-income households (**31.5%** → **27.6%**), testing 4 explanations and identifying job instability and alternative-credit use as key drivers.
- Segmented households into 4 personas using **K-Means**; deployed the analysis via Streamlit.

US Labor Market Analytics GitHub

End-to-end pipeline analyzing US employment, wage, and AI-exposure trends across 830+ occupations (2015–2025).
- Built an **ETL pipeline** in **Python and SQL** integrating 4 data sources, processing **8,900+ records** across **12 PostgreSQL tables**.
- Developed a **linear regression model** to forecast employment trends (2030–2035), benchmarking against BLS projections across **755 occupations**.
- Created an **O*NET-based AI-exposure score** and applied **K-Means clustering** to segment 755 occupations into **5 labor-market tiers**; containerized the pipeline with **Docker** and deployed it on **AWS EC2** with **S3 storage**.

Work Experience

Machine Learning Intern – COE AI Lab, VJTI Mumbai June'25 – Jul'25
- Built and trained a **CNN**-based plant disease detection model using **TensorFlow/Keras**, achieving approximately **90% training accuracy and 88% validation accuracy** on the PlantVillage dataset.
- Compared training and validation results to check for **overfitting** and generalization, improving the reliability of disease detection.
- Focused on lightweight AI solutions suitable for resource-constrained agricultural environments.